MIL-P-50472A (AR)
1 MARCH 1978
SUPERSEDING
MIL-P-50472 (MU)
20 March 1970

MILITARY SPECIFICATION

PROPELLANT M26A1, FOR USE IN CHARGE PROPELLING M189

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M26A1 Propellant for use in Charge Propelling M189 for 152MM Cartridge.

1.2 Classification. The propellant shall be M26A1 Type I, multiperforated grains.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS (see 6.11)

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9266944 - Cartridge, 152MM TP-T M411A3
9207817 - Charge Propelling, 152MM, X189
10551938 - Projectile, 152MM, TP-T M411A1 Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation, Parts 100-199

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision)).

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

(Application for copies should be addressed to the Publications and Reproduction Agency, Letterkenny Army Depot, Chambersburg, Pa. 17201).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M26A1, conforming to the requirements of MIL-STD-652 and Drawing 9207817. Total graphite (glaze and composition) shall not exceed 0.55 percent. The total volatile content shall not exceed 0.50 percent.

3.2 Ballistic assessment requirements. The propellant, when fired in the M81 gun with the projectile M411 and the cartridge case (M205) (see 6.15) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.5.1.1. At 70°F (21.1°C) and standard weapon conditions (see 6.13), the propellant shall be capable of assessment to a velocity of 2240 feet per second (fps) (678.7 meters per second (mps)) with an expected average pressure not to exceed 38,400 pounds per square inch

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(psi) (265.0 megapascals (MPa)) and a charge weight not to exceed 97 ounces (oz.) (246.4 grams (gm)).

3.3 Ballistic uniformity requirements. The propellant, when fired in the M81 gun at the established charge weight and with the projectile (M411 and the cartridge case (M205)) (see 6.15) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.5.1.2.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F (21.1°C) shall not exceed 8.0 (fps) (2.4 mps).

3.3.2 Average pressure. At standard weapon conditions, the propellant shall not produce an average pressure greater than 38,400 psi (265.0 MPa) at 70°F (21.1°C).

3.3.3 Permissible individual maximum pressure (PIMP). The propellant shall not produce a chamber pressure greater than 46,000 psi (317.4 MPa) at 125°F (51.6°C) when fired in a standard weapon (see 6.13), as applicable.

3.4 Data requirements. The contractor shall generate data, in accordance with the requirements of the data item descriptions cited in 4.4.2.1 and 4.4.2.2, documenting specific laboratory and Proving Ground test data.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Preproduction Sample Inspection and Quality Conformance Inspection shall be conducted on this item.

4.3 Preproduction sample

4.3.1 Submission. The contractor shall submit a preproduction sample as designed by the Contracting Officer for

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evaluation in accordance with 4.3.2. The preproduction sample shall consist of three (3) pilot lots of 500 lbs (1100 kg) to 1000 lbs (2200 kg) each. A new Sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity, loadability, or ballistic performance as determined by the Technical Agency (see 6.14).

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. A pilot lot which meets the requirements of the applicable drawings and specifications may be designated by the Contracting Officer as the accepted preproduction Sample. Until the preproduction Sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.4.2 Examination and testing

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652. The contractor shall submit a Propellant Description Sheet in report form and in computer-compatible format (keypunched cards) in accordance with Data Item Description DI-T-4904, giving the detailed test results (see 6.1.2).

4.4.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652, except that ten (10) containers shall be sampled for initial production lots and subsequent production lots. The Proving Ground shall submit an acceptance Test Summary in report form and in computer-compatible format (keypunched cards) in accordance with Data Item Description DI-T-4904, giving the detailed test results (see 6.1.2).

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4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify DRDAR-QAR-R and DRSAR-QAM.

4.4.2.4 Velocity uniformity (70°F) (21.1°C) (see 3.3.1). If the product of the factor (see 6.3) times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply retest criteria in accordance with 4.5.1.3.

4.4.2.5 Average pressure (see 3.3.2). If any average pressure obtained during initial or subsequent production at 70°F (21.1°C) exceeds the applicable requirement, apply retest criteria in accordance with 4.5.1.3.

4.4.2.6 Permissible individual maximum pressure (initial production). If any individual pressure observed during the 125°F (51.6°C) uniformity series, when corrected to standard conditions and recommended charge weight, exceeds the applicable requirement, the lot shall be rejected.

4.4.2.7 Permissible individual maximum pressure (subsequent production). If any individual pressure observed during the 70°F (21.1°C) uniformity series, when corrected to standard conditions and recommended charge weight, exceeds 38,400 psi (265.0 MPa), and calibration round performance is acceptable (see 4.5.1.4), the lot shall be tested at +125°F (51.6°C) in accordance with 4.5.1.2.1. If any individual pressure at 125°F (51.6°C), when corrected to standard conditions and recommended charge weight, exceeds 46,000 psi (317.4 MPa), testing shall cease and the lot shall be rejected.

4.5 Test methods and procedures

4.5.1 Proving ground assessments. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. The following tolerances shall apply to all -50°F (-45.6°C), 125°F (51.6°C) and 70°F (21.1°C) conditioning: -50°F (-45.6°C) from -50°F (-45.6°C) to -45°F (-42.8°C); 125°F (51.6°C) from 120°F (49°C) to 125°F (51.6°C); 70°F (21.1°C) from 67.5°F (19.7°C) to 72.5°F (22.5°C).

4.5.1.1 Charge establishment

4.5.1.1.1 Charge slope method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F (21.1°C).

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Three calibration rounds shall also be conditioned and fired at 70°F (21.1°C), prior to the test rounds. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.5.1.2 Uniformity

4.5.1.2.1 Initial production. Thirty rounds shall be assembled with the established charge weight of propellant. Ten rounds each shall be conditioned to -50°F (-45.6°C), 70°F (21.1°C) and 125°F (51.6°C) (as indicated by thermocoupled control round). Rounds fired at 70°F (21.1°C) shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.2), to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight) and for presence of pressure gauges. The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F (21.1°C) and 125°F (51.6°C). The average of the coefficients for ten lots shall be utilized to calculate a 70°F (21.1°C) maximum average pressure for use in subsequent production.

4.5.1.2.2 Subsequent production. Ten (10) rounds shall be assembled and tested at 70°F (21.1°C), fired alternately with calibration rounds. The means and standard deviations

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of the velocity and pressure shall be calculated on results corrected to standard conditions and recommended charge weight. If any individual pressure exceeds the 70°F (21.1°C) maximum average pressure (see 4.4.2.7), ten additional test rounds shall be conditioned and fired at 125°F (51.6°C). All results at 125°F (51.6°C) shall be corrected to standard projectile weight (see 6.15) and recommend charge weight.

4.5.1.3 Retest criteria (see Figure 1). If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary (see 6.8).

a. Test lot fails velocity standard deviation or average pressure at 70°F (21.1°C) and calibration rounds fail one or more requirements stated in 4.5.1.4, retest for uniformity.

b. Test lot fails velocity standard deviation or average pressure at 70°F (21.1°C) and any extreme test value is an outlier, retest for uniformity (see 6.9).

c. The final adjustment in charge weight exceeds one percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.5.1.1.1.

4.5.1.4 Referral criteria (see Figure 1). A test lot shall be referred to U.S. Army R&D Command, ATTN: DRDAR-QAR-R for disposition whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds one percent of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration results utilized in calculations is less than specified.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions.

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078 shall apply.

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the course of the contract, this authority is vested in another

6.1.2 Contract data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable Data Item Description
4.4.2.1	Propellant Description Sheet	DI-T-4904
4.4.2.1	Keypunched Cards	DI-T-4904
4.4.2.2	Acceptance Test Summary	DI-T-4904
4.4.2.2	Keypunched Cards	DI-T-4904

One copy each of the Propellant Description Sheet and the Acceptance Test Summary shall be forwarded to:

Commander
U.S. Army Armament Research and Development Command
ATTN: DRDAR-QAR
DRDAR-LCU-EP
Dover, New Jersey 07801

One copy of each set of keypunched cards shall be forwarded, ATTN: DRDAR-QAR-R. (Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.4 Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Throughout production, if 3 consecutive lots fail on first test or a lapse in production of 90 days or more occurs, testing shall revert back to initial production plan (see 4.5.1.2.1).

6.5 Surveillance sample. Samples shall be forwarded to U.S. Army R&D Command, Dover, New Jersey 07801, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

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6.6 Test rounds. Test rounds shall be assembled in accordance with drawing 9266944.

6.7 Proving ground test summary (see Table I).

6.8 Retest. All original and retest data shall be submitted to U.S. Army R&D Command, ATTN: DRDAR-QAR-R, Dover, New Jersey 07801.

6.9 Outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_C) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C.

6.10 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.12 PIMP basis. The PIMP value for 125°F (51.6°C) cited is based upon the upper pressure limit of the weapon.

6.13 Standard weapon. Standard weapon and standard weapon conditions refer to the weapon model and conditions of the test at the time the master calibration series was fired.

6.14 Preproduction sample. The preproduction sample is intended to provide the contractor with the necessary latitude

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to establish the propellant granulation needed to meet the required ballistic parameters. The Technical Agency and the Contractor should jointly select the one pilot lot which best represents the desired parameters. All future production lots should be of the same basic granulation as the accepted pilot lot.

6.15 Projectile weight. The average projectile weight for the cartridge in which this propellant will be loaded will be determined by the loading plant and provided for information by DRSAR-QAM to the Proving Ground. The actual average projectile weight will be used for charge weight assessment of propellant. All projectiles intended for use in test rounds shall have a weight tolerance of ± .01 lb. (4.5 gm). The Loading Authorization for the propellant lot should be annotated to indicate that the service charge weight listed is to be used only when loading with projectiles having the listed weight. If the projectile weight is different from that listed, the lot should be reassessed to the new projectile weight or the charge weight adjusted using a known projectile weight/charge weight curve. Calibration projectile weights shall be in accordance with Dwg. 10551938 except that a tolerance of ± .01 lb. (4.5 gm) shall apply. When the reference projectile lot must be replaced with a new reference lot, the new lot should reflect an average weight which is representative of current production.

6.16 Reference sample. Whenever a propellant lot is selected for reference use, a 5-lb sample shall be forwarded to U.S. Army Armament R&D Command, Dover, New Jersey 07801, ATTN: DRDAR-QAR-R for use in evaluating the propellant stockpile.

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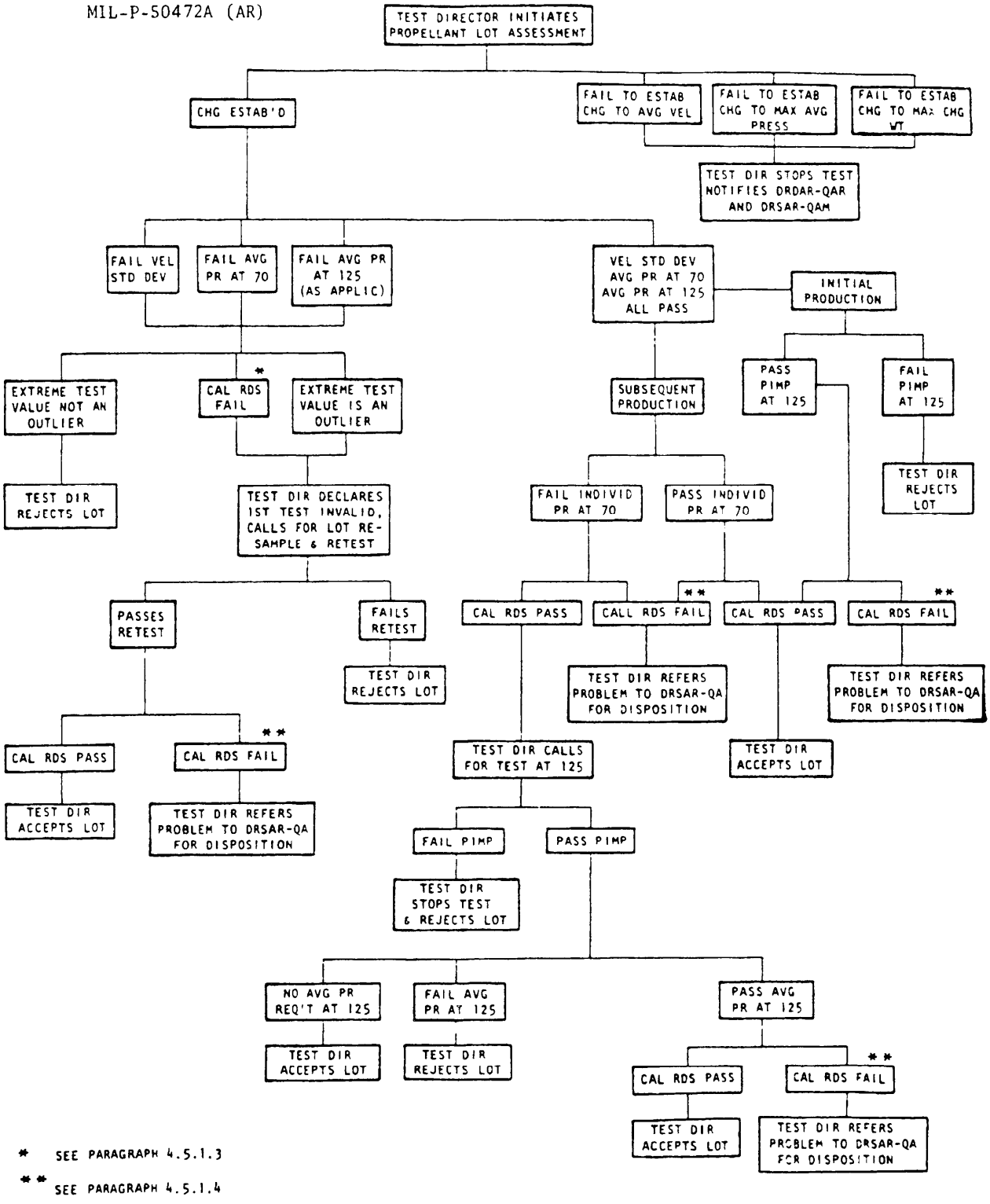
TABLE I

PROVING GROUND TEST SUMMARY

SAMPLE SIZE				TEST RESULTS							
Temperature (°F)	(°C)	Cal.	Test	Average Vel fps	mps	Std Dev of Vel fps	mps	PIMP psi	MPa	Maximum Average Pressure psi	MPa
				CHARGE ESTABLISHMENT							
70	21.1	3	9	2240	678.7	N/A		N/A		38,400	265.0
				UNIFORMITY (INITIAL PRODUCTION)							
-50	-45.6	N/A	10	N/A		N/A		N/A		N/A	
70	21.1	10	10	2240	678.7	802.4		N/A		38,400	265.0
125	51.6	N/A	10	N/A		N/A		46,000	317.4	N/A	
				UNIFORMITY (SUBSEQUENT PRODUCTION)							
70	21.1	10	10					see 4.5.1.2.2			
125*	51.6*	N/A	10*	N/A		N/A					

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.5.1.2.2).

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* SEE PARAGRAPH 4.5.1.3

** SEE PARAGRAPH 4.5.1.4

FIGURE 1 DECISION PROCESS FOR PROPELLANT ASSESSMENT

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